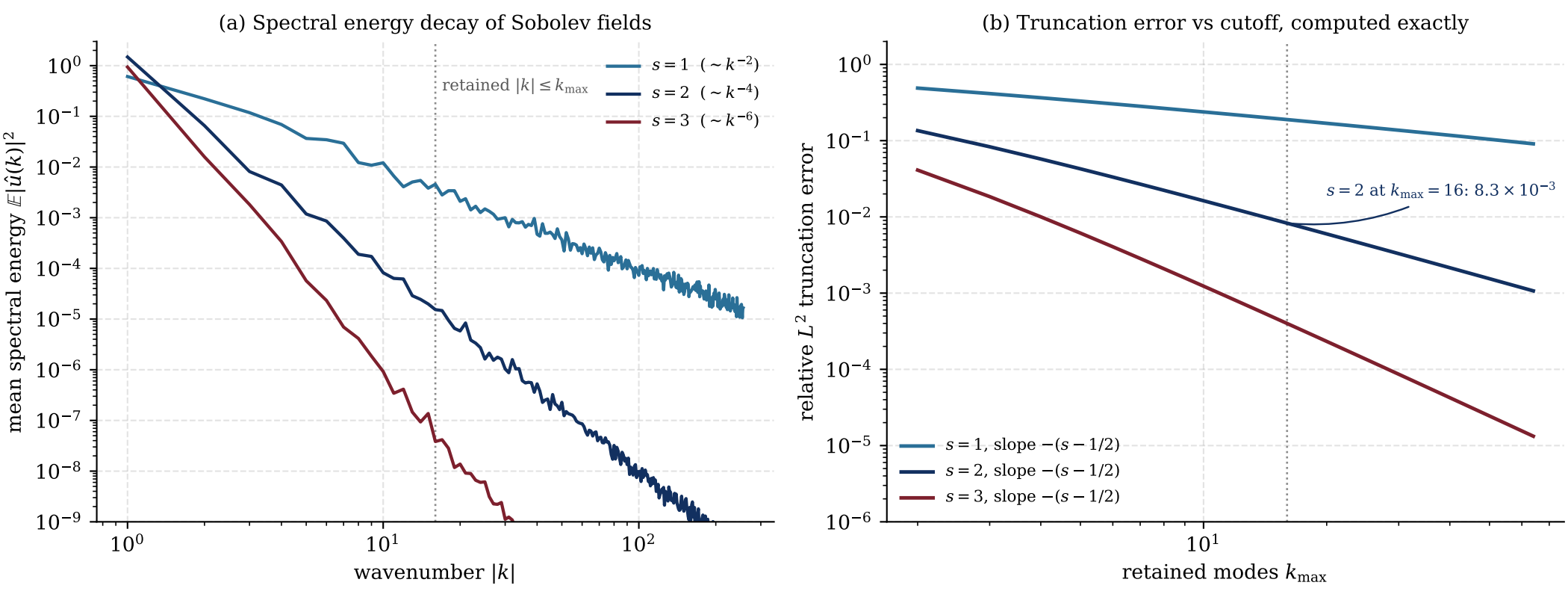
(a) Spectral energy decay of Sobolev fields
mean spectral energy $\mathbb{E}|\hat{u}(k)|^2$
wavenumber $|k|$
retained $|k| \le k_{\max}$
$s = 1$ $(\sim k^{-2})$
$s = 2$ $(\sim k^{-4})$
$s = 3$ $(\sim k^{-6})$
(b) Truncation error vs cutoff, computed exactly
relative L^2 truncation error
retained modes $k_{\max}$
$s = 2$ at $k_{\max} = 16$: 8.3×10^{-3}
$s = 1$, slope $-(s - 1/2)$
$s = 2$, slope $-(s - 1/2)$
$s = 3$, slope $-(s - 1/2)$